IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2013

EEE/ISE PART III/IV: MEng, BEng and ACGI

COMMUNICATION SYSTEMS

Monday 14 January 2013, 10.00 am

Time allowed: 3:00 hours

There are 27 questions on this paper.

Answer ALL questions.

The multiple choice questions together account for 40% of the marks.

Answers to multiple choice questions 1-20 should be given on the paper itself.

Students are not permitted to use more than one answer book.

Students are not permitted to take the question paper away.

The following are provided:
A table of Fourier transforms
A Gaussian Tail Function graph

Examiners responsible:

First Marker(s): A. Manikas
Second Marker(s): B. Clerckx

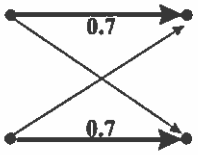
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PART-I

1. If one binary source and two binary channels are connected in cascade as shown below



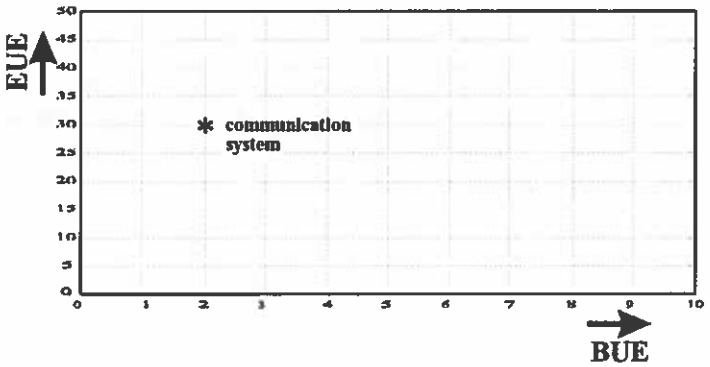
where both channels have the following forward transition probability diagram



then bir-error-rate p_e at the output of the second channel is [2 marks]

(a) 0.21;
(b) 0.28;
(c) 0.42; [x]
(d) 0.58;
(e) none of the above.

2. A digital communication system, operating at 100 kbits/sec in the presence of additive white Gaussian noise of power spectral density $\text{PSD}_n(f) = \frac{N_0}{2}$, is represented in the energy utilization efficiency (EUE) – bandwidth utilization efficiency plane (BUE), as follows:



The capacity C of the channel in bits/sec is [2 marks]

(a) 200 kbits/sec;
(b) 400 kbits/sec;
(c) 600 kbits/sec;
(d) 800 kbits/sec; [x]
(e) none of the above.

7. A FH/SSS has a data rate of 4800 bits/second and uses 4 hops per data bit. What is the bandwidth if there are 1024 hopping frequencies? [2 marks]

   (a) 19.2 kHz.
   (b) 4.096 MHz.
   (c) 4.915 MHz.
   (d) 19.661 MHz. [x]
   (e) None of the above.

8. The minimum chip rate needed by a DS-BPSK spread spectrum system to resolve a multipath, with an additional path length of 30m compared to the direct path, is [2 marks]

   (a) 10 Mchips/second; [x]
   (b) 20 Mchips/second;
   (c) 40 Mchips/second;
   (d) 60 Mchips/second;
   (e) none of the above.

9. If B_D denotes the doppler spread, B_{coh} represents the coherence bandwidth and T_c is the chip period, then in a frequency selective fast fading CDMA channel which of the following is correct? [2 marks]

   (a) $B_D > B_{\text{coh}}$.
   (b) $B_D < B_{\text{coh}}$.
   (c) $T_c > \frac{1}{B_{\text{coh}}}$.
   (d) $T_c > \frac{1}{B_D}$.
   (e) None of the above. [x]

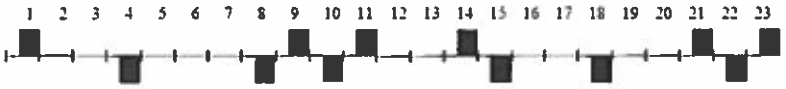
10. A binary PSK signal is decoded coherently in the presence of additive white Gaussian noise having a double-sided power spectral density 0.5×10^{-6} W/Hz. If the average signal power is 1 Watt, then the highest bit rate at which a probability of error of 10^{-5} can be achieved is [2 marks]

    (a) $\approx$ 110 kbits/sec; [x]
    (b) $\approx$ 220 kbits/sec;
    (c) $\approx$ 330 kbiys/sec;
    (d) $\approx$ 440 kbits/sec;
    (e) none of the above.

11. If D denotes the dimensionality of the signal space for an M-ary communication system which of the following statements is correct? [2 marks]

    (a) $D = 2$ for a BPSK communication system.
    (b) $D = 4$ for a QPSK communication system.
    (c) $D = 1$ for an M-ASK communication system with $M = 4$. [x]
    (d) $D = 2$ for an M-ASK communication system with $M = 4$.
    (e) None of the above.

14. The following HDB3 encoded signal



represents the binary sequence: [2 marks]

(a) 1 0 0 1 0 0 0 0 1 1 1 0 0 0 1 0 0 0 0 0 1 1 1.
(b) 1 0 0 1 0 0 0 1 1 1 1 0 0 1 1 0 0 1 0 0 1 1 1.
(c) 1 0 0 1 0 0 0 0 1 1 0 0 0 0 0 0 0 0 0 0 1 1 1. [x]
(d) 1 0 0 1 0 0 0 0 1 1 0 0 0 0 1 0 0 0 0 0 1 1 1.
(e) None of the above.

15. Consider the bipolar line code waveform $b(t) = \sum_{n=-\infty}^{\infty} \mathrm{a}[n].c(t - nT_{cs})$ where the autocorrelation function of the code sequence $\{\mathrm{a}(n)\}$ is as follows:

$$R_{\mathrm{aa}}(k) = \begin{cases} 1/2 & \text{if } k = 0 \\ -1/4 & \text{if } k = 1 \\ 0 & \text{if } k \geq 2 \end{cases}$$

If $\Pr(0) = \Pr(\pm 1) = 0.5$ and $c(t) = \mathrm{rect}\left\{\frac{t}{T_{cs}}\right\}$ then the power spectral density of b(t) is: [2 marks]

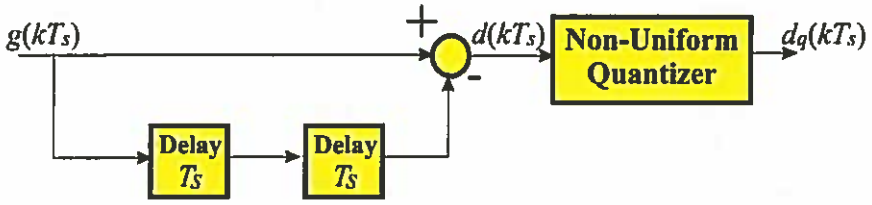
(a) $T_{cs}.\mathrm{sinc}^2\{fT_{cs}\}$;
(b) $T_{cs}^2.\mathrm{sinc}^2\{fT_{cs}\}$;
(c) $T_{cs}.\mathrm{sinc}^2\{fT_{cs}\}.\sin^2(\pi T_{cs})$;
(d) $T_{cs}^2.\mathrm{sinc}^2\{fT_{cs}\}.\sin^2(\pi T_{cs})$;
(e) none of the above. [x]

16. An analogue message signal $g(t)$ with amplitude probablity density function $0.5\Lambda\left(\frac{g}{2}\right)$ and a bandwidth of 10kHz, is applied to an A-Law /8-bit PCM system with $A = 87.6$. For this system the Signal-to-Quntization-Noise ratio (SNR_q) is [2 marks]

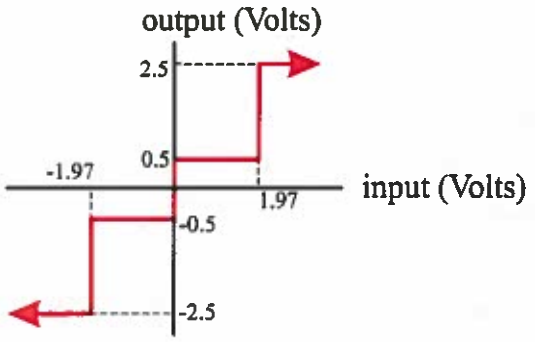
(a) 38.0059 dB;
(b) 44.9885 dB; [x]
(c) 52.77 dB;
(d) 56.52 dB;
(e) none of the above.

PART-II

21. Consider a non-white zero-mean Gaussian random source $g(t)$ having an autocorrelation function $R_{gg}(\tau) = \exp(-6000 \mid \tau \mid)$. The signal is sampled at a rate of 12k samples/sec and then is applied at the input of the differential quantizer shown below



where T_s is the sample time, k is an integer. The transfer function of the quantizer is shown below:



Find

(a) the DC level of the signal $d(kT_s)$; [1 marks]

(b) the power of the signal $d(kT_s)$. [4 marks]

Solution

$d(kT_s)$ is a Gaussian signal i.e. $\mathrm{N}(\mu_d, \sigma_d)$

(a) mean: $\mu_d = E\{d(kT_s)\} = E\{g(kT_s) - g((k-2)T_s)\} = 0 - 0 = 0$

(b) *std*: $\sigma_d = \sqrt{P_d}$
where $P_d = E\{d^2(kT_s)\} = E\left\{(g(kT_s) - g((k-2)T_s))^2\right\} =$
$= E\{g^2(kT_s)\} + E\{g^2((k-2)T_s)\} - 2E\{g(kT_s)g((k-2)T_s)\}$
$= R_{gg}(0) + R_{gg}(0) - 2R_{gg}(2T_s) = 2R_{gg}(0) - 2R_{gg}(2T_s)$where $T_s = \frac{1}{12k}$
$= 2 \times 1 - 2 \times \exp(-1)$
$= 1.264$
i.e. pdf=$\mathrm{N}(0, \sqrt{1.264})$

23. Consider two analogue signals x(t) and y(t). The signal x(t) has a uniform amplitude probability density function, i.e.

$$\text{pdf}_x(x) = \frac{1}{3}\text{rect}\left\{\frac{x}{3}\right\}$$

while the signal y(t) has a Gaussian amplitude probability density function $\text{pdf}_y(y)$ of mean μ_y and standard deviation equal to $\sqrt{P_x}$ where P_x denotes the average power of x(t). Find [10 marks]

(a) the average power P_x of the signal x(t);

(b) the differential entropy H(x) of x(t);

(c) the differential entropy H(y) of y(t);

(d) the entropy power of x(t);

(e) the entropy power of y(t).

Solution

(a) average power:

$$P_x = \mathcal{E}\left\{x^2(t)\right\} = \int_{-\infty}^{\infty} x^2.\text{pdf}_x(x).dx = \int_{-3/2}^{3/2} x^2.\tfrac{1}{3}.dx = \tfrac{3}{4} = 0.75$$

(b) diff entropy: $\text{H}_x = -\int_{-\infty}^{\infty}\text{pdf}_x(x).\log_2\left\{pdf_x(x)\right\}dx$

$$= -\int_{-3/2}^{3/2}\tfrac{1}{3}.\log_2\tfrac{1}{3}dx = \log_2 3 = 1.585$$

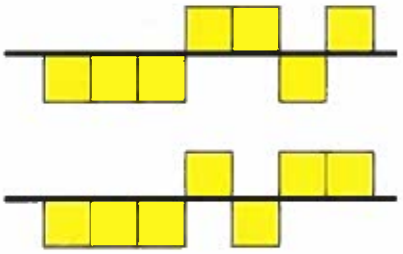
(c) diff entropy of y(t) source:

$$H_y = \log_2\sqrt{2\pi e P_x} = \log_2\sqrt{2\pi e \times 0.750} = 1.8396$$

(d) entropy power of x(t)$=\frac{1}{2\pi e}2^{2\text{H}_x} = \frac{1}{2\pi e}2^{2\times 1.585} = 0.52698$

(e) entropy power of y(t)$=P_x=0.75$

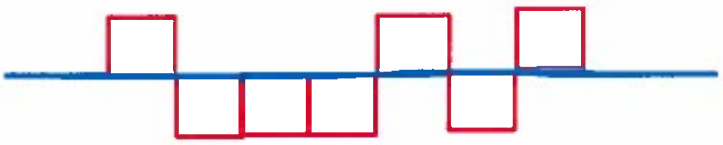
25. Two m-sequence PN-signals, generated by two 3-stage shift registers, are shown (one period per signal) below.



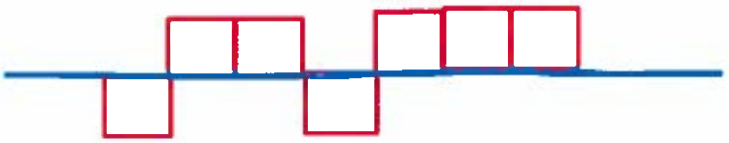
Construct a Gold code signal from these two PN-signals by delaying the second signal by one chip period. [5 marks]

Solution

The 2nd PN-signal delayed by one chip period



Gold code (mod-2 addition)



27. Consider a digital cellular DS-BPSK CDMA communication system which employs three directional antennas each having 120° beamwidth, thereby dividing each cell into 3 sectors. The system can support up to 201 users/subscribers and operates with a data bit-rate of 500 kbits/sec in the presence of additive white Gaussian noise of double-sided power spectral density 10^{-9}. With a bit-error-probability for each user of 3×10^{-5}, a power equal to 10 mWatts, and a voice activity factor $\alpha = 0.375$, find:

(a) the average energy per bit E_b; [5 marks]

(b) the equivalent EUE (EUE_{equ}); [5 marks]

(c) the processing gain (PG) of the system. [5 marks]

Solution

(a) $P = 10mW$
$r_b = 500\,\mathrm{kbits/\ sec} \Rightarrow T_{cs} = \frac{1}{500}$ msec
$\mathrm{K} = 201$ users
$\mathrm{No} = 2{\times}10^{-9}$
$p_e = 3 \times 10^{-5}$
$a = 0.375$
$s = 1/3$
$E_b = PT_{cs} = 10 \times 10^{-3} \times \frac{1}{500} \times 10^{-3} = 2 \times 10^{-8}$

(b) $p_e = \mathrm{T}\{\sqrt{2\,\mathrm{EUE}_{equ}}\} \Rightarrow 3 \times 10^{-5} = \mathrm{T}\{\sqrt{2\,\mathrm{EUE}_{equ}}\}$
$\Rightarrow$ (using "tail" graph supplied)
$4 = \sqrt{2\,\mathrm{EUE}_{equ}} \Rightarrow \mathrm{EUE}_{equ} = 8$

(c) However, $\mathrm{EUE}_{equ} = \frac{E_b}{\mathrm{N_o} + \mathrm{N_j}}$
where $E_b = PT_{cs}$ and $\mathrm{N_j} = \frac{(\mathrm{K}-1).P.a.s}{B_{ss}} = \frac{(\mathrm{K}-1).P.a.s}{\mathrm{PG}/T_{cs}}$
Therefore, $\mathrm{EUE}_{equ} = \frac{PT_{cs}}{\mathrm{N_o} + \frac{(\mathrm{K}-1).P.a.s}{\mathrm{PG}/T_{cs}}} \Rightarrow \ldots\ldots\ldots \Rightarrow \mathrm{PG} = \frac{(\mathrm{K}-1).P.a.s.T_{cs}}{\frac{PT_{cs}}{\mathrm{EUE}_{equ}} - \mathrm{N_o}}$
$\Rightarrow \ldots \Rightarrow \mathrm{PG} = 1000$